

Core stability

by PAUL HODGES

Core strength and stability is very important to me. Tennis is all about rotation of the body and my ability to create power. I incorporate a lot of abdominal, back and glute exercises into my gym sessions.

Samantha Stosur

INTRODUCTION

'Core stability' refers to control of proximal regions of the body during movements and function. Although the concept appears straightforward, different interpretations or misconceptions of the term, and differences in applying this concept to the design of exercise programs for lumbopelvic pain, have led to substantial confusion and debate.

'Stability' has been interpreted to mean many things and, commonly, a simple objective to *restrict* motion. This has led to the development of many exercise programs that emphasise static maintenance of a specific lumbopelvic position during loading. Although this may be appropriate for some functions, restriction of motion does not necessarily assure stability of a system.

A stable system is one in which its intended *position* or *trajectory* (i.e. movement path) can be maintained, despite perturbation or disturbance.¹ This may be achieved by restricting motion in some tasks (e.g. restriction of spinal motion during lifting), but requires movement to manage, reduce or transfer internal and external forces in others (e.g. counter-rotation of the shoulders and pelvis during walking) (Fig. 11.1). Further, in a complex dynamic system such as the spine, the stability of many goals (e.g. spine control, breathing, balance) must be maintained concurrently.²

The 'core' refers to proximal regions of the body, which can include the spine, pelvis, scapulothoracic region and even the hips and shoulders. These regions are functionally distinct from the distal limbs, which have a critical role in fine controlled, goal-directed movements. The principal functions of the core are to provide a foundation for limb movements, a stable base for limb muscle contraction and a major contribution to overall postural control of the

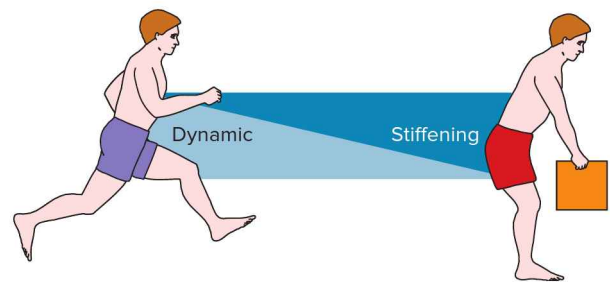


Figure 11.1 Optimal control of the spine depends on a balance between movement and stiffness. At one end of the spectrum are functions such as lifting that require high stiffness of the trunk; at the other are functions such as running and walking that demand movement to meet the demands of the task. Both too much and too little movement and stiffness could underlie suboptimal loading

body. This latter aspect is particularly important as the trunk contributes about 70% of body weight. The focus of this chapter is consideration of the lumbopelvic region, but clinical management must involve consideration of the adjacent regions, as they cannot act independently.

When the concepts of 'stability' and the 'core' are fully appreciated, it is clear that the design of exercise programs for lumbopelvic pain require the term 'core stability' to be applied in its broadest sense to encompass all features that might be necessary to ensure optimal control. This requires a motor control or learning approach, whereby muscle activation patterns, postures or alignments, and movements are trained, tuned or optimised to meet the demands of function³ (Fig. 11.2). Although restriction of motion may be encouraged for some functions and some patients, this

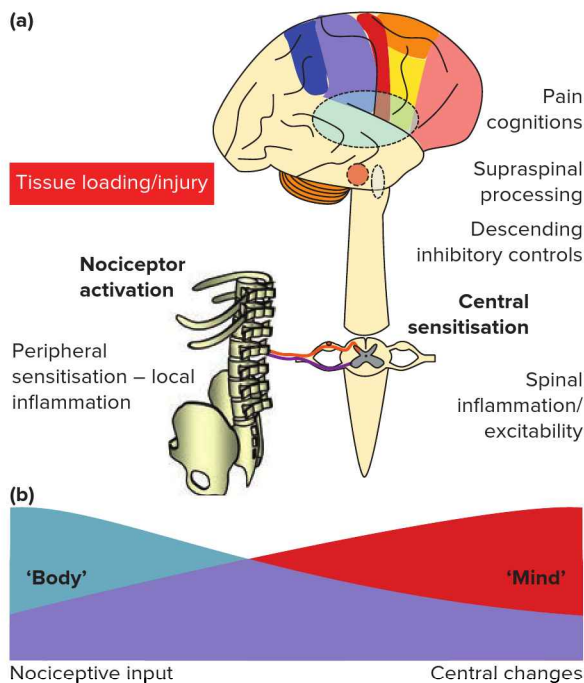


Figure 11.2 Contribution of body and mind to persistent pain. (a) The experience of pain involves peripheral and central components. Individual patients will vary in the relative contribution of peripheral and central contributions to their pain presentation. Some will maintain a major contribution of ongoing nociceptive input from suboptimal loading of tissues; others will have little input from tissues and major contribution from central processes. Motor control training is likely to be most effective for individuals with an ongoing contribution from peripheral afferents. (b) Different treatments will be required along the spectrum from treatments that target the body such as motor control training for core stability at one end, to psychologically grounded treatments for the mind at the other end

depends on the presentation of the individual patient, whereas for other functions and other patients the exercise may aim to increase motion. 'Motor control training' is a general term used to describe a comprehensive approach that uses a range of strategies (from low load training of precise control of muscles, posture and movement; progression of exercise with increasing load and complexity; and functional retraining; through to strength and endurance training, as required) in combination with education and behavioural strategies to promote pain management.

Training must:

- optimise the balance between movement and stiffness (which differs between tasks)
- consider the coordination of the multiple functions of the trunk and of the trunk muscles

- individualise the program to match the patient's unique presentation.

This chapter provides an overview of the rationale for motor control training to optimise core stability in the management of lumbopelvic pain, outlines key principles underlying this approach and discusses common misconceptions. It also describes the typical process for planning a motor control training program; from detailed assessment of motor control, formulation of the clinical hypothesis, development of an exercise program to optimise loading, and planning for progression of exercise.

RATIONALE FOR MOTOR CONTROL TRAINING FOR LUMBOPELVIC DYSFUNCTION

The premise underlying the effectiveness of motor control training to optimise core stability is that for many individuals with lumbopelvic pain, suboptimal loading of structures of the spine and pelvis (as a result of the muscle activation, posture or alignment, and movement typically adopted by an individual) places the tissues at risk of trauma or microtrauma, and drives peripheral nociceptive input which contributes to the development and persistence of pain (Fig. 11.2).

Motor control training of core stability aims to optimise tissue loading to prevent further injury and to reduce nociceptive afferent discharge.³ Although it is acknowledged that symptoms may persist in the absence of ongoing nociceptive input, and it is widely agreed that central sensitisation will contribute to the pain experience for many⁴ (Fig. 11.2), individuals who retain a peripheral component to their symptoms are likely to benefit from reduction of this input through optimisation of the manner in which the individual loads their tissues and body.

With the link between loading and pain in mind, it is clear that the exercise must be individualised to modify the features that are specific to the individual's unique presentation. Examples of suboptimal loading that could cause ongoing nociceptive input include reduced muscle activation leading to insufficient constraint or control of joint motion; sustained alignment of the pelvis and spine at the end of range of motion placing excess load on passive tissues;⁵ excessive motion of the spine and pelvis rather than the hip during leg movement;⁶ and excessive compressive load on the spine from augmented co-contraction of trunk muscles (e.g. during lifting⁷). Motor control training aims to change the suboptimal features of muscle activation, posture and movement to achieve optimal core stability with optimal tissue loading.

Although it would be impossible to test the validity of every permutation of suboptimal tissue loading, evidence is beginning to emerge that loading can be related to symptoms,

and optimisation of loading may be necessary for recovery. For instance, training that aims to restore activation of the deep abdominal muscle, transversus abdominis, is more effective for individuals found to have poor activation of this muscle on clinical assessment at baseline, and recovery of symptoms is related to recovery of function of this muscle in those individuals.^{8,9} Further, low back symptoms are reduced when spine position is controlled during hip rotation in individuals who present with early and excessive lumbopelvic motion during this task.¹⁰

A thoughtful clinical reasoning approach is essential to:

- determine whether the patient's pain presentation includes a component that is maintained by peripheral nociceptive input
- identify any features of muscle activation, posture/alignment and movement that underpin the suboptimal tissue loading for the individual through detailed clinical assessments
- develop a clinical hypothesis to account for the relationship and inform treatment planning
- develop an exercise program to target the motor control changes to optimise tissue loading.

Of course it is critical to recognise that the presentation of lumbopelvic pain generally involves features within multiple domains of the biopsychosocial model (Chapters 15 and 23). In addition to peripheral nociceptive input from abnormal tissue loading, there may also be sensitisation of pain processing at multiple levels of the nervous system and some features of unhealthy pain cognitions and social features (Fig. 11.2). Depending on the individual patient's presentation, treatment is likely to be optimised when all factors relevant for the patient's presentation are considered in treatment design. This will include considering additional measures to manage attitudes and beliefs about pain, as well as other aspects such as pain-coping skills.

MOTOR CONTROL TRAINING TO OPTIMISE CORE STABILITY— KEY PRINCIPLES AND COMMON MISCONCEPTIONS

Key principles that underlie the application of motor control training to lumbopelvic pain are reviewed below to clarify the basis for clinical decisions when designing an exercise program to optimise core stability. A number of misconceptions regarding the approach also require clarification.

Optimal motor control requires a balance between movement and stiffness

The primary focus of many exercise programs for core stability has been to restrict motion (e.g. exercises that target maintenance of lumbar and pelvic position during limb movement or side-bridging). Although these

strategies to brace the trunk are appropriate and necessary for some tasks (e.g. lifting a 100 kg weight from the floor, which requires high stiffness of the spine), they are not appropriate for other functions, such as walking and running, where movement is essential for optimal task performance. Most functions require a balance between movement and stiffness to optimise tissue loading; the optimal balance depends on the task being performed.

Sufficient stiffness is necessary to prevent collapse and maintain alignment. Movement is required to absorb impact, transfer load, minimise energy expenditure, allow some variation to share load between related structures, and to move along a trajectory. Even static tasks generally involve some movement (e.g. standing involves small movements to compensate for a range of challenges, including the cyclical postural disturbance from breathing¹¹). A healthy system has the capacity to shift along a continuum from high stiffness at one end and high mobility at the other to match the balance between movement and stiffness required by task demands (Fig. 11.1). Individuals with lumbopelvic pain often present with inaccuracy in matching task demands.

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Evidence is emerging for poor control across this stiffness-mobility spectrum in people with lumbopelvic pain. This can present as excessive motion,¹² excessive stiffness¹³ or elements of both.

A tennis player may use excessive lumbar rather than hip motion during a racket swing. A runner may brace the spine with excessive muscle co-contraction, and limit shock absorption and load transfer through the trunk. There may be a change in threshold, where a strategy normally reserved for a high load task is used at low load. All of these can lead to excessive tissue loading which is problematic for function, not only in terms of suboptimal lumbopelvic control, but also for the broader functions of the trunk. For instance, balance in standing is compromised if the trunk is stiffened.¹⁴

On this basis it would seem inappropriate to apply exercise universally to enhance stiffness (e.g. bracing co-contraction of abdominal and extensor muscles) for every patient or for every functional context. Such strategies should be reserved to enhance stiffness for moments of high demand. Other exercise is required to train appropriate movement.

Optimal lumbopelvic control involves three main neural strategies

The nervous system employs three basic strategies to optimise lumbopelvic control. The first is reactive control in which the nervous system activates a pattern of muscle activation in response to sensory input.¹⁵ This class of

response is involved in formulating the response when the system is unexpectedly disturbed (e.g. trip, slip, push) or to fine tune muscle activation.

The second is preparatory control when the muscle activation required to control the lumbopelvic region can be initiated in advance of a disturbance.¹⁶ This type of adjustment can only be used when demands can be predicted (e.g. voluntary initiated stepping, catching a watched ball).

As reactions take time and not all disturbances can be predicted, the system relies on a third strategy of ongoing tonic muscle activation at a low percentage of contraction to maintain a state of preparedness, just in case the spine is perturbed.¹⁷

All muscles of the trunk can contribute to these three classes of motor control strategy. Dysfunctions may present in each and it may be necessary to consider all three in a treatment program (e.g. train reactive control through challenging the system with balance tasks; train preparatory control by pre-activating muscles in advance of a task; train sustained gentle tonic activation).

Optimal motor control requires a whole system, not a single muscle

A single muscle cannot be responsible for optimal motor control and core stability, and dysfunction would be unlikely to involve a single muscle. Instead control must depend on the integrated function of a whole system of muscles, and the muscles cannot be considered independently of the alignment of the spine and pelvis, and the pattern of movement.

All muscles of the lumbopelvic region contribute to optimal control of the region and this is influenced by muscles of adjacent regions (e.g. hip/thorax). But this does not mean that all muscles serve the same role. As a general principle, deeper muscles with attachments to individual vertebra and limited moment arms for torque generation can play a primary role in controlling intersegmental motion. The larger more superficial muscles with a greater moment arm for torque generation are more critical for stiffening the spine when co-contracting in antagonist muscle pairs, and for generation and control of movement when acting asymmetrically or in specific patterns of activation that promote torque in a specific direction.

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Research evidence universally shows that the transversus abdominis and multifidus muscles are not the only muscles required for lumbopelvic control and no function involves activation of these muscles in the absence of activation of other trunk muscles.

Both of these contributions are essential and deficits in either would lead to problems.

There has been considerable debate regarding the function and dysfunction of the transversus abdominis and multifidus muscles.^{18, 19} Although always part of a more complex interplay of muscle activation, these muscles often act differently to other trunk muscles (e.g. less dependent on force direction, early and tonically),^{16, 20} which is related to their distinct mechanical contribution to control.

Transversus abdominis acts primarily through generating intra-abdominal pressure and tension of fascial structures, both of which appear to control intersegmental and pelvic motion in multiple directions. Multifidus is situated close to the centre of rotation of the lumbar segments, suggesting limited role in torque generation. These muscles commonly have structural and functional changes in people with pain.²¹⁻²⁴

Together these observations provide a foundation for detailed consideration of these muscles in the management of lumbopelvic dysfunction. Backed by biomechanical and modelling data of a contribution to optimal control, individuals with compromised activation of these muscles are likely to contribute to abnormal tissue loading. These patients would be likely to benefit from restoration of control as part of a comprehensive training program that includes considering other muscles, posture and movement.

Motor control training involves rehabilitation of whole system

Considering the breadth of features of motor function that can underpin suboptimal tissue loading, it is clear that a motor control training program must address multiple features, including muscle activation, posture, or alignment and movement (Fig. 11.3). The exact features addressed and the combination depends on the individual's presentation and function. Thus, to optimise core stability, planning a training program to optimise motor control requires detailed assessment to identify features related to abnormal loading.

In terms of muscle activation, motor control of core stability would be suboptimal if muscle activity was insufficient, excessive or lacked coordination in terms of timing, amplitude or pattern of activation. Insufficient activation (e.g. delayed activation of transversus abdominis²¹) may allow excessive motion or poor quality of motion within the normal range of motion, both of which would lead to increased tissue load.

Excessive activation (e.g. augmented activation of obliquus externus abdominis) could restrict motion (limiting shock absorption and compromising function); increase joint load; and reduce variation (reduce sharing of load between structures). Poor coordination of muscle

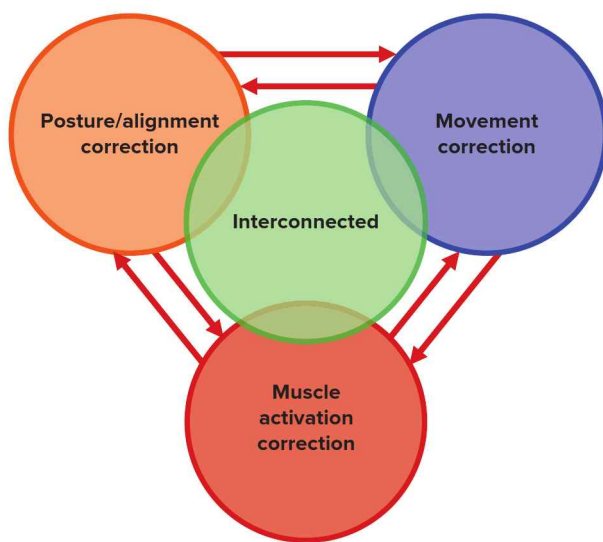


Figure 11.3 Suboptimal motor control generally involves features of posture or alignment, movement and muscle activation. The combination of features differs between individuals and each needs to be addressed, although there will be some interaction between them with exercise that targets one aspect having an effect on others

activation (e.g. *en bloc* activation of back muscles rather than differential activation) could change the location and amount of tissue load. None of these observations is universal and it would be illogical to target any change universally with exercise. The exercise program would involve the selection of techniques and strategies to change the pattern of activation.

A common clinical strategy that requires additional explanation is the common objective to isolate the activation of deeper muscles with specific exercises. It is important to clarify that this strategy does not aim to encourage a patient to activate *only* these muscles in function, but to train the *skill* of activating the muscles so that a patient can then integrate their activation into function, along with correcting any other suboptimal feature of motor control that is identified in the patient's presentation.

Like all elements of core stability, it is necessary to find a balance, and more muscle activity is not necessarily better. Although some degree of muscle activity, intra-abdominal pressure and fascial tension improves control,²⁵ if these are in excess of demands, this will lead to additional load or restriction of movement. It is necessary to tune muscle activity to meet, but not exceed, demands and to reduce it if the muscle is overactive.

Alignment or posture is also important. Although there is not one posture of the lumbopelvic region that is ideal for every patient, there is a range of features

that clearly increase load on the lumbopelvic region.⁵ Examples include resting at the end of range (e.g. sitting with posterior pelvic tilt with lumbar flexion); hanging on passive tissues (e.g. resting on an adducted hip in standing to tense the iliotibial band); and active holding of a posture by excessive muscle activation or asymmetrical postures. Changes to alignment or posture to reduce the resulting tissue loading may be necessary to optimise loading. This could be achieved with multiple techniques such as instruction, feedback, taping and manual therapy (see below).

It is critical to consider the pattern of movement. Like posture, there are features of movement that are likely to be suboptimal.²⁶ Every movement involves more or less ideal solutions and there are many ways that it can be suboptimal. Some examples include too little movement of a region or segment; too much movement; too little variation; and poor relationship between adjacent regions or segments. Again, selecting the strategy to change movement strategy in order to optimise load is required.

It is clear that motor control training for core stability requires an individualised approach. This is based on clinical and experimental data. When outcomes of randomised clinical trials are considered, there is a tendency for greater clinical effect size in trials that involve individualised training of specific subgroups of patients.²⁷ The least impressive outcomes are from trials that apply a generic treatment to a group of patients with non-specific back pain.

From an experimental perspective, there is clear evidence that exposure to pain induces adaptations that vary between individuals. For instance, if healthy individuals are provoked by a noxious stimulus to the back, all adapt to protect the spine, but no two participants achieve this with identical patterns of muscle activation.²⁸ Further, not all patients present with deficits in activation of the deep trunk muscles (transversus abdominis and multifidus muscles), and those that do have a deficit in these muscles have a better response to training that includes training of these components than do patients who do not have a deficit to start with.⁹ Individualisation of training requires detailed assessment.

Motor control training involves a motor learning approach

When the objective of training is to modify specific features of motor control to optimise core stability, this can be targeted with a motor learning approach. Skill learning commences with identifying and then correcting the fault or suboptimal feature of a task. Once the modified skill is learnt, this can then be integrated into functions of increasing complexity. This applies to the learning of many skills.²⁹

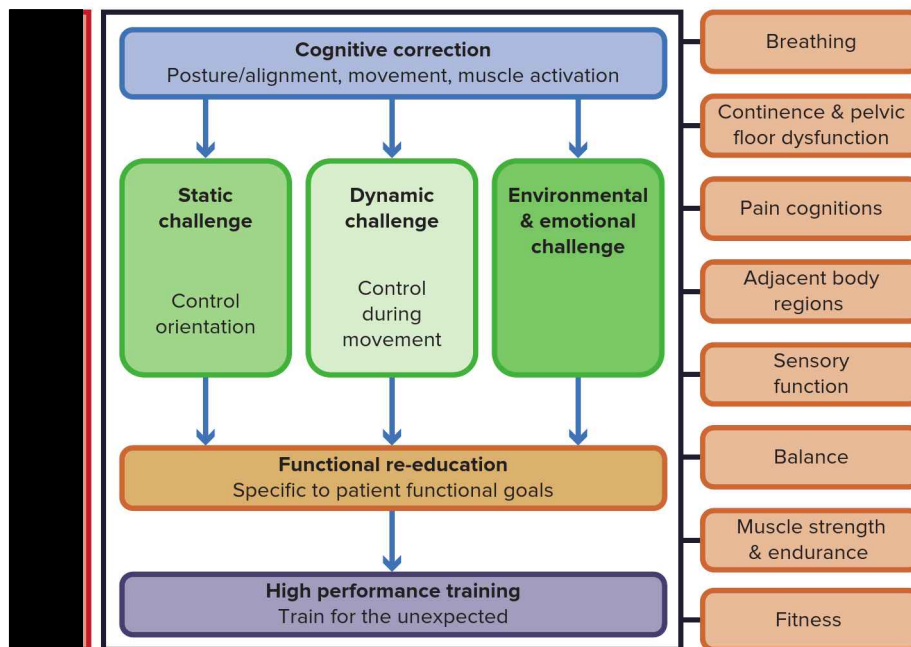


Figure 11.4 Motor control training for core stability. Training begins with detailed training of features of posture or alignment, movement and muscle activation that are considered to contribute to suboptimal tissue loading. Once these skills are learnt it is necessary to introduce static and dynamic challenges to encourage refinement and diversification of the skill. Functional training is necessary to ensure modified motor control strategies are introduced into functional tasks. Training then progresses to high performance for the unexpected. A range of issues, highlighted in boxes to the right, can be barriers to recovery and require consideration for individual patients if they are present

When applied to motor control training for core stability (Fig. 11.4), the initial stage would involve modifying and tuning the features of muscle activation, posture or alignment, and movement that are identified in the assessment. This can be achieved by conscious, cognitive correction of the feature, with feedback (visual, ultrasound imaging, palpation, electromyography), or the use of techniques such as taping and manual guidance (see below).

Skill learning can be enhanced by simplifying the task (e.g. non weight-bearing positions, slower speed) and segmenting a complex task into component parts (e.g. practice of correction of a muscle activation, practice of a specific alignment) prior to practice of a complete task. The selection of which features to address first depends on the individual. It is perhaps best guided by the features that are likely to achieve the greatest change in pain provocation or most able to be achieved by the patient.

As with any skill training, once the skill is learnt the patient can be progressed by challenging the patient with more complex tasks (e.g. increased load, more challenging postures and movements). This can involve progressive levels of increased demand and it is generally necessary

to consider static control (of a specific alignment of the spine and pelvis) or dynamic control (of the lumbopelvic region as it moves) or control of the lumbopelvic region during whole body movement. Finally, the functions that a patient indicates are problematic can be practised with correction of the features of motor control.

Interplay between motor control and biology of pain

Motor control training would be expected to have a greater effect for patients with a nociceptive component. As highlighted above, but critical to reiterate, motor control training aims to optimise the load on the tissues and this is most likely to impact an individual's symptoms if they continue to have a nociceptive contribution to their symptoms (Fig. 11.2). Patients whose pain is maintained by central sensitisation may derive benefit from education of healthy movement, but this will impact their function through mechanisms other than reduction of peripheral nociceptive input, such as experience with healthy movement as a step towards fear-deconditioning.

It must be remembered that pain is not 'in' the tissues and is a result of interplay between nociceptive input,

biological processes in the periphery and the nervous system that both inhibit and facilitate the pain experience, and cognitive and emotional aspects such as distress or fear (Fig. 11.2). Pain is a construct of the nervous system on the basis of each of those inputs and processes. Pain can be maintained by ongoing nociceptive input from the tissues and in its absence.

As highlighted above, the role of optimisation of tissue loading (through training of core stability) in the management of pain will be likely to be greatest in individuals with ongoing nociceptive contribution to pain. This type of pain is considered to have recognisable features such as a clear relationship with movement that is predictable and proportional, localised to an area, has clear aggravating and easing factors, and is localised on palpation.³⁰

A recent clinical trial showed that motor control training and a graded activity general exercise approach were equally effective when the whole group of non-specific low back pain patients were considered together; however, motor control exercise was more effective for those who responded to a questionnaire that, although developed to identify instability (which may be considered too narrow a view of suboptimal motor control), also identifies characteristics that are consistent with nociceptive-type pain.³¹

Many patients also present with a host of biological changes (in addition to those that relate to suboptimal loading, such as modified inflammatory system activity³²), and psychosocial features that also impact their pain presentation.³³ When present, these are likely to impact the efficacy of an exercise approach focused on motor control. For some, those processes are dominant and should be the primary target of intervention, for others, a combined approach may be helpful—restoration of function, but with attention to how people move.

An important consideration is that behavioural, psychologically informed treatments often encourage a patient to ignore pain, yet correction of motor control may demand consideration of when a movement is performed in a non-provocative manner. Care is required in selecting language to ensure the patient does not develop the belief that their spine is at risk and must be protected, which are concepts likely to reinforce unhealthy pain cognitions.

Implications of the role of trunk muscles in respiration and bladder and bowel function

In addition to their role in controlling the spine, trunk muscles, particularly those that surround the abdominal cavity, have critical roles in breathing and bladder and

bowel function. These functions must be coordinated with the contribution of these muscles to lumbopelvic control. Yet in the present of disorders of continence (e.g. stress urinary incontinence), breathing (e.g. asthma) and back pain, the contribution to any or all of these functions may be disrupted.³⁴ Coordination of these functions is particularly critical for application to sport and more demanding physical activity.

Breathing requires modulation of pressure in the thorax—alternating activation of the diaphragm and other chest wall muscles in inspiration, and abdominal and internal intercostal muscles in expiration. These modulations of activity are coordinated with breathing in healthy individuals, but modified in those with respiratory disease and some with low back pain. Training to coordinate functions is likely to be required.

In terms of bladder and bowel function, continence is maintained if the pressure in the urethra or rectum exceeds that in the bladder/colon. Abdominal and diaphragm muscle activity increase intra-abdominal pressure, which concomitantly raises bladder and bowel pressure. Pelvic floor muscle activation is necessary to elevate intra-abdominal pressure. Thus, coordination between spine control and continence functions of the whole lumbopelvic system, including the abdominal wall, back, diaphragm and pelvic floor, requires consideration. This is particularly the case in the presence of pelvic floor muscle dysfunction, which may present as compromised or excessive muscle activation, both of which would have consequences for optimal control of the spine.

Is motor control training effective for everyone or is it more effective when targeted to specific individuals?

As highlighted above, there is emerging evidence that motor control training for core stability is more effective for some individuals, and more effective when tailored to the individual patient. Hypothetically, motor control training would be more effective for patients and athletes presenting with symptoms that are related to modified loading (including an ongoing nociceptive component). As highlighted above, recent work highlights that the treatment is most effective for patients with a high score on a questionnaire which purports to indicate signs of instability.³¹ Further work is required to determine whether this questionnaire instead reflects signs of ongoing nociceptive contribution to pain. Further, training is more effective for patients who are identified to have a deficit in control of the deep trunk muscles at baseline.^{8, 9} It seems logical that this is not a treatment that can be applied in a blanket manner, but applied to the right patients and in a tailored manner.

PRINCIPLES OF THE CLINICAL APPLICATION OF MOTOR CONTROL TRAINING

As highlighted above, the basis of a motor control approach for core stability is to optimise the manner in which a person loads the tissue of the spine and pelvis by addressing any suboptimal features of muscle activation, posture or alignment, and movement. It follows then that the application of motor control training requires careful assessment to identify the suboptimal features, and then design a treatment or exercise program to target these features, before progressing into functions of increasing complexity. The basic stages of a program to optimise motor control of core stability are presented in

Figure 11.4. A brief overview is provided below (for a detailed review see Hodges et al.³).

Assessment of motor control for core stability

Assessment involves a series of screening tests to identify key features of muscle activation, posture and movement that could imply suboptimal loading, combined with more detailed analysis to identify the specific nature of the loading issue (Fig. 11.5). Subjective assessment is essential to provide initial guidance of key functional demands and provocative or relieving postures, movements or functions. Physical tests of motor control are based on the premise that it is possible to identify features of posture, movement or muscle

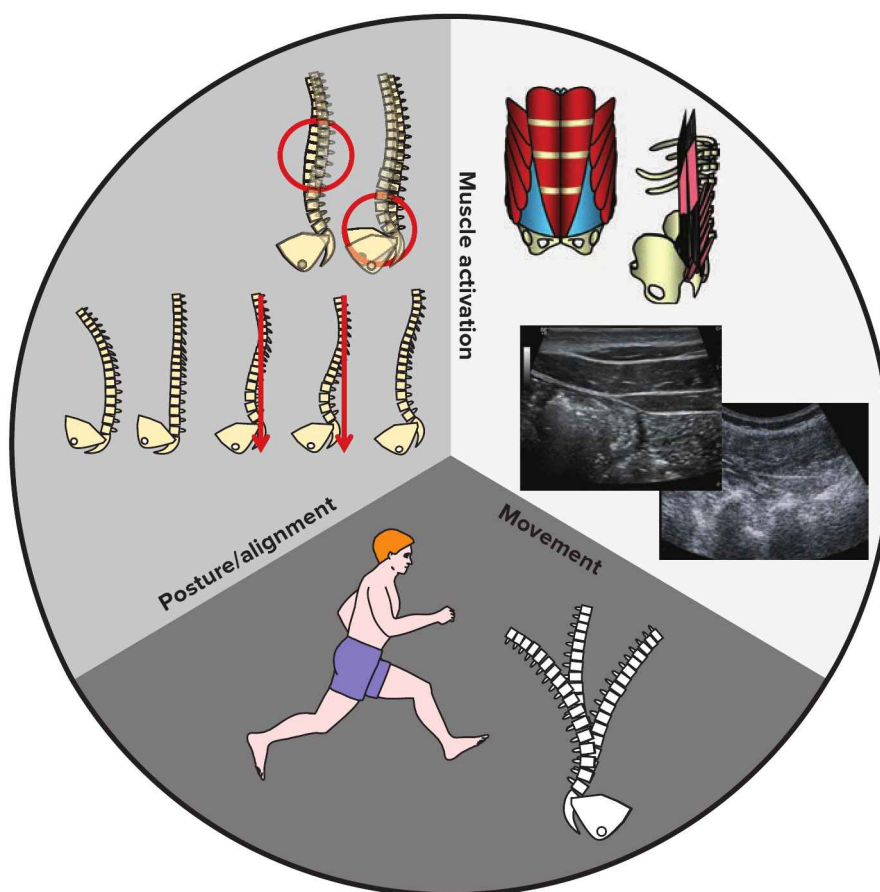


Figure 11.5 Assessment of motor control of core stability requires detailed assessment of posture or alignment, muscle activation and movement. Posture or alignment is compared against an ideal which is refined for consideration of the individual patient. Muscle activation involves assessment of underactivity, overactivity, threshold and asymmetry of the whole trunk muscle system. Assessment can be informed by evaluating the capacity to activate the deeper trunk muscles and concurrent substitution by the more superficial or global muscles. Movement assessment involves assessing the pattern of movement against an optimal strategy during physiological movements (e.g. trunk flexion), functional movements (e.g. sit-to-stand) and specific clinical movement tests (e.g. hip rotation in prone)

activation that deviate from a predicted ideal and, in many cases, provoke symptoms. Provocation or relief of symptoms does not always provide a perfect indicator of the relevance of a motor feature for a patient's presentation.

In some cases a clear relationship will be identified. But in others a highly inconsistent, unrepeatable relationship with motor features and disproportionate (area or intensity) pain provocation may be present. This may indicate a dominant central sensitisation, which might not be ideally managed with a motor control approach. For prevention of future symptoms of a patient in remission or for primary prevention, pain cannot be used as a guide, and the assessment depends on the clinician's understanding of ideal control.

Assessment of muscle activation

Assessment of muscle activation aims to identify evidence of overactivity or underactivity of any muscle(s). Observation of posture and movements can provide some indication of muscle activation (e.g. sitting with posterior tilt and extension at the thoracolumbar junction may suggest low activation of the multifidus and excessive activation of the thoracolumbar erector spinae).

A variety of clinical tests is available to analyse muscle activation. Some examples are described here. The common clinical tests of independent activation of the transversus abdominis and multifidus aim to assess both (i) the function of these deeper muscles (e.g. poor ability to activate the transversus abdominis independently is related to delayed activation in function³⁵), and (ii) evidence of substitution or overactivity of the more superficial muscles. Clinical tools, observation, palpation, ultrasound imaging and electromyography can be used for the assessment. The capacity of the more superficial trunk muscles to control the lumbopelvic region can be derived from static tests that assess the capacity to control alignment when force is applied (e.g. load-applied limb movement). The threshold load that can be controlled, asymmetry in control and excessive bracing are assessed.

Assessment of posture or alignment

Assessment of posture or alignment begins with the identification of features that differ from the predicted optimal posture. Although not optimal for all, a starting point for evaluating sitting or standing posture is to identify features that differ from an ideal that includes a level or slightly anteriorly tilted pelvis, gentle lumbar lordosis with a smooth transition to a gentle thoracic kyphosis, cervical lordosis, symmetry in frontal plane and no rotation in the transverse plane. For many, this posture optimises load sharing and requires only gentle muscle activation to maintain.³⁶ This goal needs to be modified or refined when considering the individual patient as the same posture is not ideal for all.

Assessment involves identifying features that deviate from this ideal with confirmation of the relevance for the patient's presentation by evaluating the response of pain or discomfort and muscle activation. The ideal may feel unusual to the patient, but it should not be painful or require high muscle activation to hold. Analysis of alignment is generally necessary in a range of relevant functional postures to identify suboptimal features using a similar approach.

Assessment of movement

Assessment of movement involves a similar process to assessment of posture except that every movement pattern has its own ideal pattern of motion and muscle activation. For instance, forward flexion involves sharing motion between the hips, pelvis and spine with a specific sequence. Clear understanding of the requirements of movement is required. Movement tests typically involve assessment of physiological movements (e.g. trunk flexion), standardised functional movements (e.g. sit-to-stand) and specific movement tests (e.g. control of the spine and pelvis during hip rotation in prone). Several clinical approaches have been developed that include specific tests of movement to assist in identifying relevant features of movement (e.g. movement systems impairment²⁶). In the case of the management of athletes, it is essential for the clinician to have a clear understanding of the features of optimal performance of the typical motor functions involved.

Consider subgrouping patients

Assessment can be facilitated by consideration of patient subgroups. Subgrouping based on motor control features has been developed by several groups and can help to simplify the process of identifying the cluster of features that may require attention with training.^{26, 37} Subgrouping using this approach is based on the premise that there are groups of patients who present with a similar combination of motor control features that are related to their symptoms. When a clinician can identify the subgroup into which a patient belongs, then the task of selecting the combination of features that need to be prioritised and targeted with treatment can be simplified. For instance, if a patient is identified with a 'flexion' motor control impairment,³⁷ then the therapist can predict that relevant motor control features may include control of the lumbar lordosis, poor dissociation of motion of the pelvis and lumbar spine from the thoracolumbar junction, and compromised activation of the multifidus muscles. In this way subgrouping simplifies the clinical reasoning process by aiding the pattern recognition.

Training of motor control for core stability

At the conclusion of the assessment, a clinical hypothesis is formulated regarding the likelihood that suboptimal tissue loading is relevant for the patient's condition,

Table 11.1 Application of motor learning to skill training to lumbopelvic pain**1. Cognitive phase of learning**

Learn and optimise features of basic motor skill, high degree of attention to:

- pattern of muscle activation
- alignment or posture
- movement strategy

2. Associative phase of learning

- challenge control with increasing demand
- diversify skill to different contexts
- refine skill:
 - static and dynamic challenges
 - different physical and mental contexts
 - enhance capacity for physical performance—endurance training or strength training for high-intensity control

2. Associative phase of learning

- attention demand reduced
- enhance versatility of skill:
 - integrate learned features into function
 - high performance training for the unexpected

and the relationship between specific features of motor control and symptoms or potential symptoms. On this basis, an exercise program is tailored to modify the specific features of posture or alignment, movement and muscle activation to optimise tissue loading, with a plan for progression of exercise towards improved function.

Training applies the same principles of motor learning used to develop any new skill, and involves a sequence of steps from initial learning of the movement components followed by progressions of increasing complexity and load. The initial step involves modifying the target features of motor control that have been identified in the assessment to be relevant for the patient's symptoms. This is generally achieved with cognitive training using the principles of motor learning (see Table 11.1).²⁹ This process progresses from initial optimisation of the target features of motor control using a combination of exercise and clinical strategies that are tailored to the individual, followed by increasing the challenge to consolidate and diversify the control, and finally, integration into function.

There is substantial evidence that the initial step of training should be specific to the identified problem (e.g. control of a specific alignment or muscle activation) and that cognitive training with attention to the task (feedback and understanding of the goal) leads to greater cortical changes than unskilled practice with load. The principles of segmentation (practice of the specific elements of a task that require change prior to integration into the complete task) and simplification (practice with reduced load, demand or complexity to enable high-quality performance of the trained element) are applied. A range of clinical techniques can be

tailored to the individual to achieve the required change in motor performance (Table 11.2). The basic objective of the first phase is to identify a strategy the patient or athlete can employ to modify the feature of motor control that requires improvement, followed by practice.

Once the patient or athlete has learnt to change the feature of motor control that underpins the suboptimal loading, it is then necessary to progress the individual to be able to control these features with progressively challenging tasks. This associative phase of motor learning²⁹ aims to *refine* the skill, *diversify* the skill and increase the physical capacity.

Progression should include both static and dynamic exercises. Static progressions are very common and there are many examples of exercise approaches that fit this domain. The basic premise is that the alignment of the pelvis and spine is optimised and then maintained statically while load is applied through either the limbs or an external force. Threshold loads and asymmetries are identified and targeted with training. Typical examples are pilates exercises (e.g. reformer exercise with the pelvis and spine maintained statically as the limbs move to slide the body), leg loading tasks in supine, side-bridging, etc. Exercise difficulty is progressed by increasing the load, speed or task instability (e.g. less stable surfaces), and the ability to maintain control of the alignment is the cue to determining the task difficulty that can be tolerated.

Dynamic progressions are less commonly applied in typical exercise programs, but they are no less critical for learning optimal control of the spine and pelvis. Dynamic training involves loading the corrected motor control

Table 11.2 Clinical techniques to aid motor learning

Technique	Purpose	Examples
Instructions	Verbal instructions to highlight how a task is to be performed differently	'Roll forwards on tailbone'; 'gently flatten the lower abdomen'
Imagery	Mental image to help a patient understand the modification that is required	'Lengthen spine'
Manual guidance	Therapist's hands on the patient to guide performance for patient to experience optimal performance	Therapist's hands on superior edge of sacrum to guide anterior rotation of pelvis
Manual cues	Patient's hands placed on specific landmarks to provide feedback of optimal alignment or movement	Patient's hands on manubriosternal junction and pubic symphysis to guide upright trunk alignment
Dissociation tasks	Change relative motion between adjacent regions of body	Separate motion of the lumbar spine and pelvis from the thorax
Feedback	Provide information regarding accurate or inaccurate performance of task. Used to provide information of movement strategy, alignment and muscle activation	Visual—mirror; ultrasound imaging (muscle activation) Kinaesthetic—tactile/palpation (finger and thumb on xiphoid and navel to provide feedback of thoracolumbar alignment), tape (applied to the lumbar spine for feedback of alignment) Auditory/visual—EMG biofeedback of obliquus externus abdominis to reduce overactivity
Muscle activation	Modification of muscle activity to enable change in posture or movement	Palpation, observation, EMG biofeedback
Cues/reminders	Technique to remind a patient to modify performance	Tape applied to the lumbar spine that provides input to skin when the lumbar spine flexes
Manual therapy and connective tissue techniques	Modification of joint mobility, muscle activation, muscle tension	Manual therapy to the thoracic spine to reduce thoracic paraspinal muscle activity

feature during movement. As for static training, load is applied while the patient maintains some objective, which in this case is a specific feature of movement. Dynamic training can involve control of the spine as it moves, control of the spine as the whole body moves, or control of spine while the patient is supported on an unstable surface. Examples might include control of the lumbar lordosis during rotation of the spine, and maintaining spine alignment while using movement to balance on a balance board.

Progression to functional exercise and functional tasks is essential. Transfer of improved motor control strategies to the patient's function is best achieved the closer to the function the person practices. This may require segmenting and simplifying a complex function initially to enable correct or optimal performance of the improved skill as a component of the complex task, but should progress to practice of the complete task. It is also necessary to provide challenges—physical (load,

endurance), emotional (stress, fatigue) and environmental (e.g. predictable, unpredictable) to ensure the skill is diversified to address all contexts (Fig. 11.4). Functional training requires that the patient or athlete has learnt to perceive (proprioceptive awareness) the correct performance in order to correct errors. A final phase is to challenge patients to prepare them for high performance with higher load and unexpected challenges.

All patients need to progress through the basic sequence from initial correction of the relevant features of motor control to function, but some also require consideration of other issues to ensure optimal performance is achieved and to overcome potential barriers. For an individual patient this might include management of issues of control of adjacent joints (e.g. foot control problems that modify leg, pelvis, spine movement; scapulothoracic control), continence, breathing, balance, muscle strength and endurance, cardiovascular fitness, unhealthy beliefs and cognitions about pain and injury

and so on. Identification of these issues may be apparent from subjective questioning, observation of movement and specific tests or questionnaires. Treatments may need to be tailored for these aspects and draw on the expertise of other professionals.

Motor learning requires permanence of the learnt motor function. Through practice and diversification, it is a key objective for the improved motor control strategy to be automatically applied in function. In situations of high demand, reinforcement through additional cognitive attention to the correction may be necessary. Development of a versatile and robust motor strategy requires practice and challenge.

Training of motor control for core stability for prevention of pain and injury

Although there is not yet sufficient high-quality evidence to support the assumption that applying the principles of motor control training can prevent pain and injury, it is plausible that identifying suboptimal features of posture or alignment, muscle activation and movement in an individual, and modifying those features with training, will help to prevent future development of pain. This is a topic of considerable interest and programs have been developed and applied with this in mind. The challenge is to identify suboptimal features that may be relevant for future development of pain or injury when provocation of symptoms is not available for confirmation that relevant features are identified. This approach depends on theoretical models that predict which features may be relevant and how they can be recognised. For athletic and sports performance this requires detailed understanding of optimal motor control for the specific sporting tasks. The approach of screening motor control for risk is strongly advocated by some.²⁶

Considerations for training of motor control for core stability in athletes

Motor control of core stability is particularly relevant for athletes at all levels. The major issues that require specific consideration are:

1. the features of motor control required to complete the task (e.g. the specific combination of movements, postures and muscle activation), which may be highly refined for the completion of the sport-specific function
2. the features of the task that may promote suboptimal control
3. training load and demands required to maintain sporting or athletic performance. This demands that the clinician has expert knowledge of the sport being considered.

Running is an ideal example to illustrate the breadth of goals that are inherent in optimal core stability for an athlete. When running, it is essential to maintain upright posture relative to gravity, to maintain the relative alignment between adjacent segments, both large (body regions—thorax relative to lumbar spine relative to pelvis) and small (intervertebral segments); while moving the spine to transfer load between body regions, absorb reactive moments from foot contact, minimise energy expenditure, achieve progression of the body through space; and maintaining respiration, continence, balance and cardiovascular function.

Any perturbation to this system can disrupt the stability of the system (e.g. landing on the outside of the foot, an unexpected increase in respiratory demand, a modified muscle activation or movement strategy to avoid pain) and lead to interference with other elements in this delicate balance. A holistic view that considers the interaction between elements is necessary. For instance, a runner with low back pain may have augmented co-contraction of trunk muscles to protect the spine from provocation of pain, but this may then compromise the efficiency of the spine for shock absorption, load transfer and use of the chest wall motion to breathe. Detailed understanding of the mechanics of running (Chapter 8) is critical to identifying features that are suboptimal.

CONCLUSION

This chapter aimed to discuss the rationale for motor control training for core stability, to dispel some common myths and to provide an overview of the comprehensive program required to optimise motor control. It is critical to remember that core stability does not imply stiffness with a sole objective to restrict motion. Optimal control depends on a balance between movement and stiffness that is matched to the demands of the task. Motor control training must be matched to the presentation of the patient and to his or her functional demands.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e